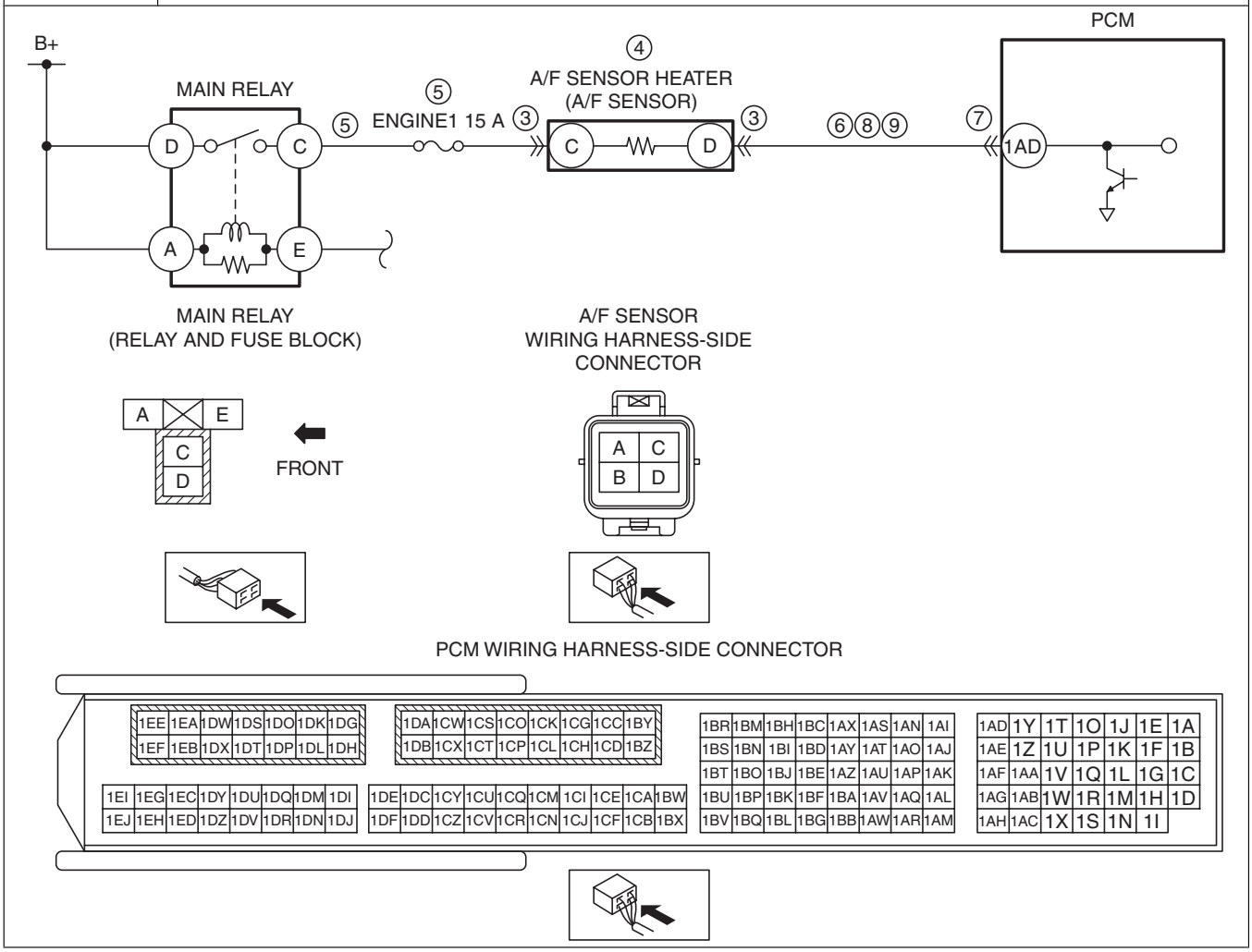
DTC P0030:00 [SKYACTIV-D 1.5]

id0102q2700300

DTC P0030:00	A/F sensor heater control circuit range/performance problem
DETECTION CONDITION	• With the following conditions met, a condition is detected by the PCM in which the voltage while the A/F sensor heater is operating or not operating is out of the specified value **for 5 s**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — A/F sensor heater control duty value: **10—90 %** **Diagnostic support note** • This is an intermittent monitor (A/F sensor). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• A/F sensor connector or terminals malfunction • A/F sensor heater malfunction • Short to ground or open circuit in A/F sensor heater power supply circuit — Short to ground in wiring harness between ENGINE1 15 A fuse and A/F sensor terminal C — ENGINE1 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and A/F sensor terminal C • Short to ground in wiring harness between A/F sensor terminal D and PCM terminal 1AD • PCM connector or terminals malfunction • Short to power supply in wiring harness between A/F sensor terminal D and PCM terminal 1AD • Open circuit in wiring harness between A/F sensor terminal D and PCM terminal 1AD • PCM malfunction

DTC P0030:00	A/F sensor heater control circuit range/performance problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD FREEZE FRAME DATA/SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (A/F sensor related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT A/F SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the A/F sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT A/F SENSOR HEATER** • Inspect the A/F sensor heater. (See AIR FUEL RATIO (A/F) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the A/F sensor, then go to Step 10. (See AIR FUEL RATIO (A/F) SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**INSPECT A/F SENSOR HEATER POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the A/F sensor connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the A/F sensor terminal C (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE1 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE1 15 A fuse and A/F sensor terminal C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and A/F sensor terminal C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
6	**INSPECT A/F SENSOR HEATER CONTROL CIRCUIT FOR SHORT TO GROUND** • Verify that the A/F sensor connector is disconnected. • Switch the ignition off. • Inspect for continuity between A/F sensor terminal D (wiring harness-side) and body ground. • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between A/F sensor terminal D and PCM terminal 1AD. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 10.
		No	Go to the next step.
7	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**INSPECT A/F SENSOR HEATER CONTROL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the A/F sensor and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the A/F sensor terminal D (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between A/F sensor terminal D and PCM terminal 1AD. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 10.
9	**INSPECT A/F SENSOR HEATER CONTROL CIRCUIT FOR OPEN CIRCUIT** • Verify that the A/F sensor and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between A/F sensor terminal D (wiring harness-side) and PCM terminal 1AD (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between A/F sensor terminal D and PCM terminal 1AD. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Drive the vehicle on a flat road surface 5 times under the following conditions. 1. Idle the engine for 20 s. 2. Shift to 2nd gear and rapidly accelerate the vehicle to **35 km/h {22 mph}** and drive the vehicle at a constant speed for **25 s**. 3. Stop the vehicle. 4. Switch the ignition to off and wait for **20 s or more**. 5. Shift to 3rd gear and rapidly accelerate the vehicle to **50 km/h {32 mph}** and drive the vehicle at a constant speed for **50 s**. 6. Idle the engine for **20 s**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.